Assignment 3 Report - Docker

Student: Tina Nguyen
Course: CS 5525 - Cloud Computing

Step 1: Build Docker Image

```
Command Prompt

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C:\Users\nguye>cd "C:\Users\nguye\OneDrive\Documents\assignment3_tina"

C:\Users\nguye\OneDrive\Documents\assignment3_tina>dir
 Volume in drive C is Windows-SSD
 Volume Serial Number is 16BB-582E

 Directory of C:\Users\nguye\OneDrive\Documents\assignment3_tina

03/26/2026  01:37 AM    <DIR>          .
03/28/2026  12:30 AM    <DIR>          ..
03/25/2026  12:30 AM    <DIR>          .github
03/28/2026  01:52 AM                93 app.py
03/27/2026  11:36 PM                75 Dockerfile
               2 File(s)            168 bytes
               3 Dir(s)  253,444,317,184 bytes free

C:\Users\nguye\OneDrive\Documents\assignment3_tina>type app.py
print("Hello from Tina Nguyen's Docker image!")
print("This is Assignment 3 for CS 5525.")

C:\Users\nguye\OneDrive\Documents\assignment3_tina>docker build -t tina-assignment3 .
[+] Building 1.3s (9/9) FINISHED                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                     0.0s
 => => transferring dockerfile: 112B                                                     0.0s
 => [internal] load metadata for docker.io/library/python:3-alpine                       0.8s
 => [auth] library/python:pull token for registry-1.docker.io                            0.0s
 => [internal] load .dockerignore                                                        0.0s
 => => transferring context: 2B                                                          0.0s
 => [1/3] FROM docker.io/library/python:3-alpine@sha256:faee120f7885a06fcc9677922331391fa690d911c020abb9e8025ff3d  0.0s
 => => resolve docker.io/library/python:3-alpine@sha256:faee120f7885a06fcc9677922331391fa690d911c020abb9e8025ff3d  0.0s
 => [internal] load build context                                                        0.0s
 => => transferring context: 126B                                                        0.0s
 => CACHED [2/3] WORKDIR /app                                                            0.0s
 => [3/3] COPY app.py .                                                                  0.0s
 => exporting to image                                                                   0.2s
 => => exporting layers                                                                  0.1s
 => => exporting manifest sha256:88b7ce1ed6751037ffc4d02de66c7811450b19a8e89701937aeb3e00e9273432  0.0s
 => => exporting config sha256:aaf0a85e1133196edb58ec414f2c646f13c07e686608bb817e3d2ce0d1522c3c  0.0s
 => => exporting attestation manifest sha256:53dd52962ee0fb91b6e6f92a036c3ecc3c50be41408f357ccb8dfbbcc687a66e  0.0s
 => => exporting manifest list sha256:5dd787564020ca16c9e2c275e5a52ece9aef54cccfd1030168b32a5ed80290bc  0.0s
 => => naming to docker.io/library/tina-assignment3:latest                               0.0s
 => => unpacking to docker.io/library/tina-assignment3:latest                            0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ujo81li1tcixpx2ff3n9aylkd
```

Step 2: List Images and Run Container

```
C:\Users\nguye\OneDrive\Documents\assignment3_tina>docker images

                                                                                    i Info →   U  In Use

IMAGE                                  ID           DISK USAGE   CONTENT SIZE   EXTRA
docker/welcome-to-docker:latest        c4d56c24da4f       22.2MB         6.03MB    U
hello-world:latest                     452a468a4bf9       25.9kB         9.49kB    U
tina-assignment3:latest                5dd787564020       72.7MB         17.7MB

C:\Users\nguye\OneDrive\Documents\assignment3_tina>docker run --rm tina-assignment3
Hello from Tina Nguyen's Docker image!
This is Assignment 3 for CS 5525.
```

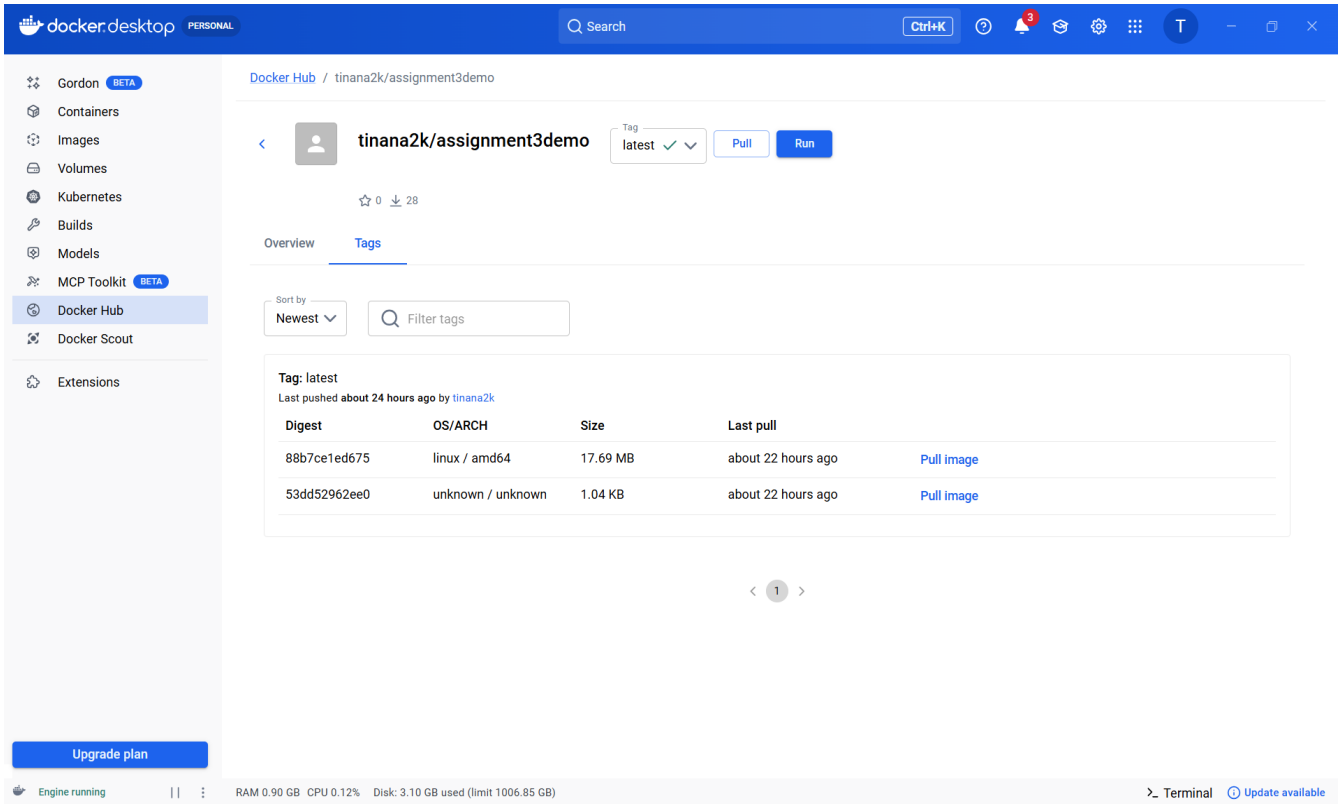
Step 3: Docker Login

```
C:\Users\nguye\OneDrive\Documents\assignment3_tina>docker login
Authenticating with existing credentials... [Username: tinana2k]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```

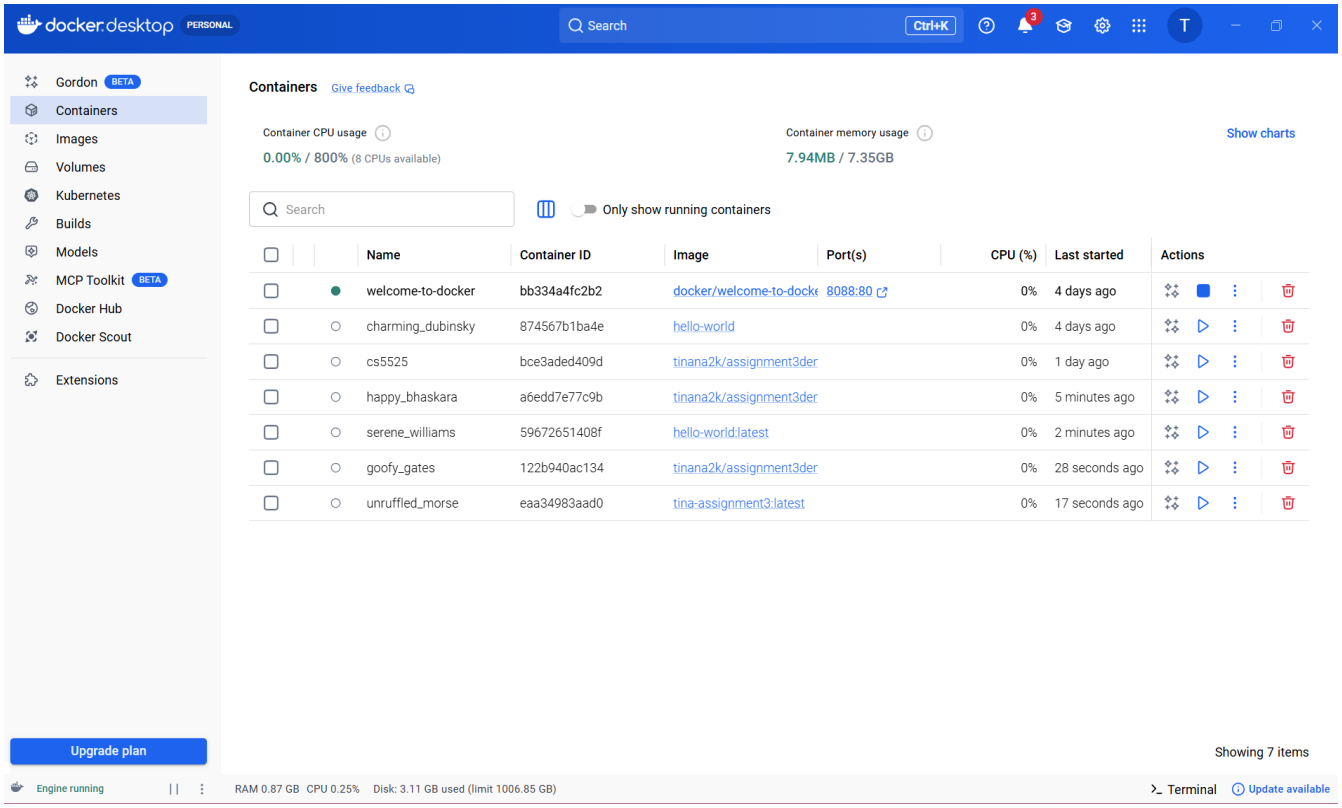
Step 4: Tag Docker Image



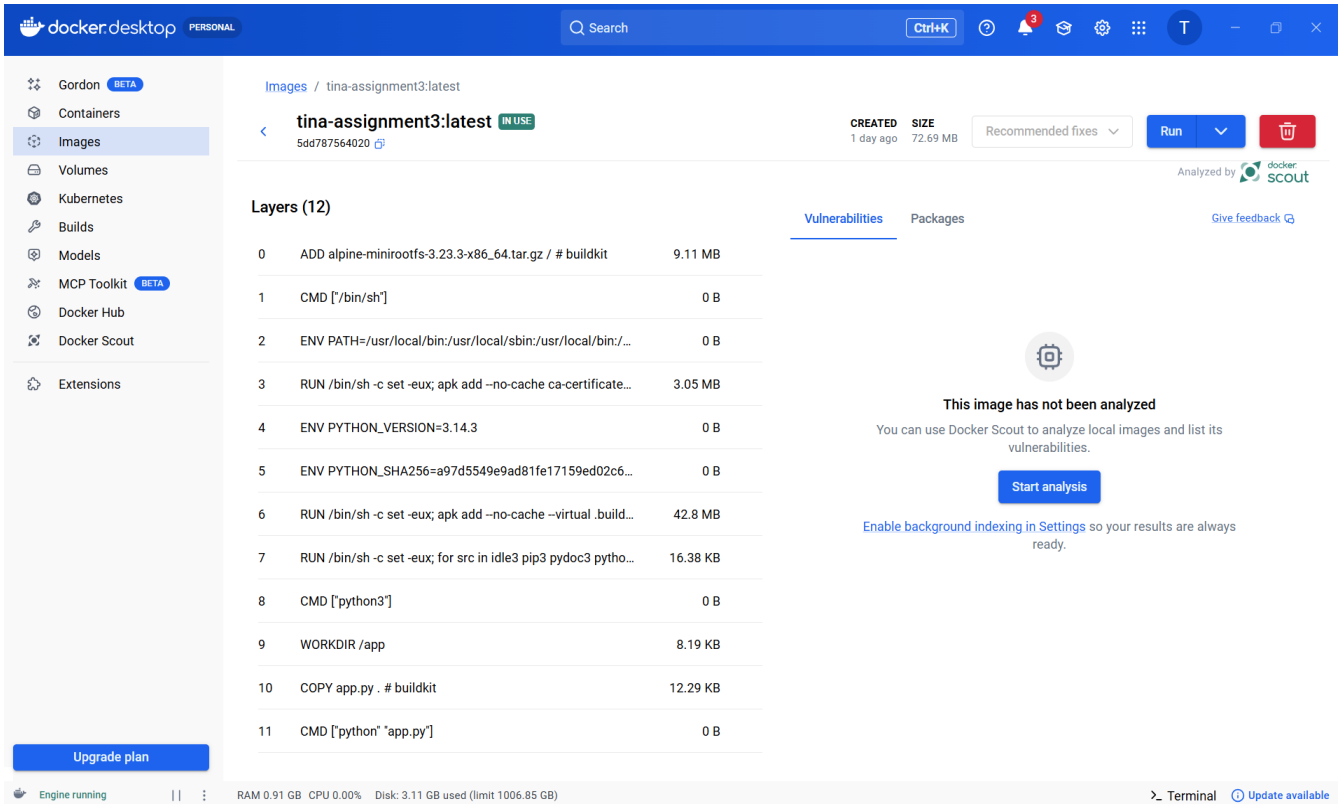
Step 5: Push Image to Docker Hub

```
C:\Users\nguye\OneDrive\Documents\assignment3_tina>docker push tinana2k/assignment3demo
Using default tag: latest
The push refers to repository [docker.io/tinana2k/assignment3demo]
78f80fdd9adb: Pushed
bc6db4519801: Pushed
0805a1082be0: Pushed
2800a7aef8b1: Pushed
64cef7c30f63: Pushed
589002ba0eae: Pushed
3566efde290b: Pushed
latest: digest: sha256:5dd787564020ca16c9e2c275e5a52ece9aef54cccfd1030168b32a5ed80290bc size: 856
```

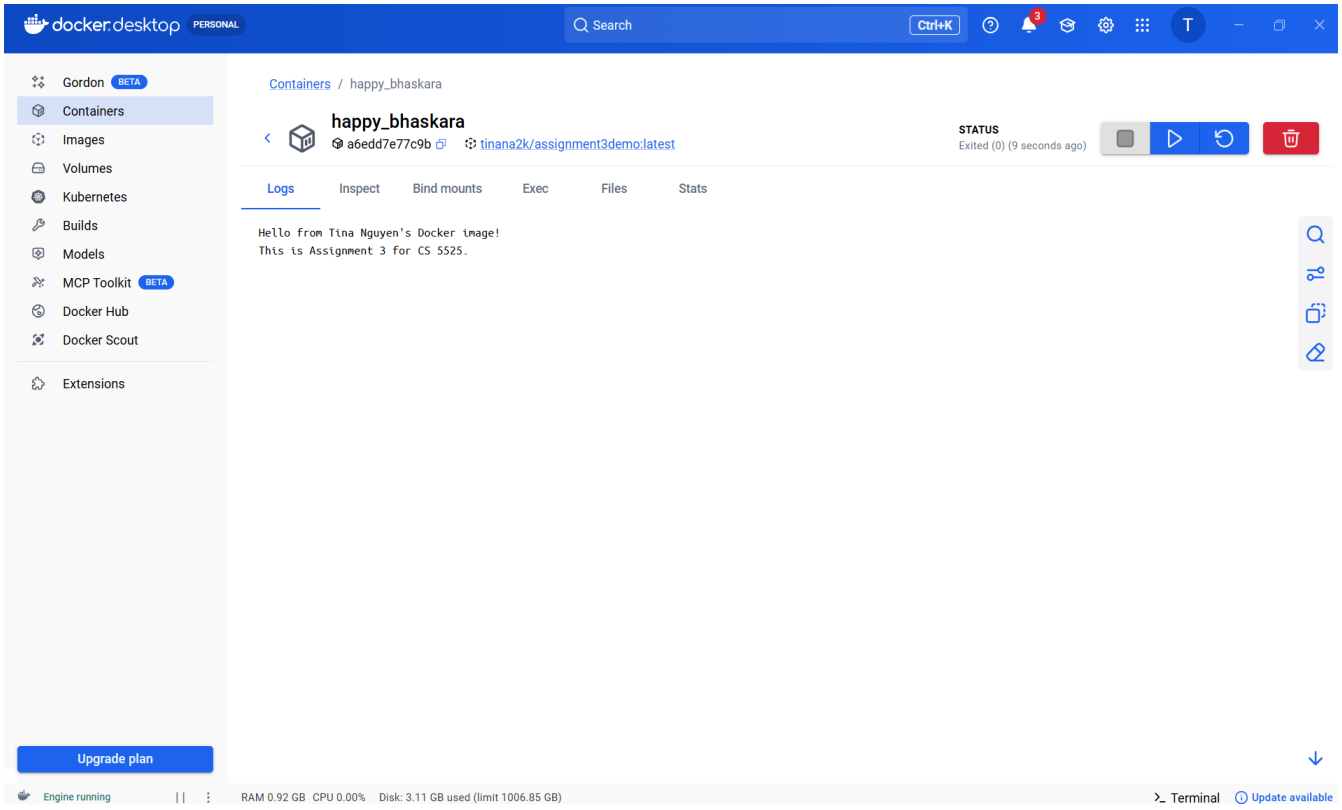
Docker Hub Repository



Docker Hub Image Details



Docker Container Output


docker desktop
PERSONAL
Search
Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Engine running
Containers / happy_bhaskara
happy_bhaskara
a6edd7e77c9b
tinana2k/assignment3demo:latest
STATUS
Exited (0) (9 seconds ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
Hello from Tina Nguyen's Docker image!
This is Assignment 3 for CS 5525.
RAM 0.92 GB CPU 0.00% Disk: 3.11 GB used (limit 1006.85 GB)
Terminal
Update available

Epilog

This assignment provided hands-on experience with Docker containerization, including building images, running containers, and deploying images to Docker Hub. One of the most valuable lessons learned was understanding the workflow from creating a simple Python application to packaging it into a container and sharing it publicly. The most challenging part was troubleshooting issues related to Dockerfile configuration and file naming conventions, especially ensuring that the Dockerfile was correctly recognized. The most enjoyable part was successfully running the container and seeing the output displayed, confirming that the Docker image was built and executed correctly. Overall, this assignment improved my understanding of containerization and its importance in modern cloud computing workflows.